

T1. 1p

Simple inverting op-amp comparator with no feedback, different from zero threshold: circuit, VTC, threshold voltage, waveforms to illustrate the operation.

T2. 1p

Class B power amplifier: circuit; operation for positive and negative input voltage; VTC; waveforms for input and output voltages highlighting crossover distortions; waveforms for output voltage, collector current and collector-emitter voltage. Assume a sinusoidal regime with maximum amplitude of the output voltage.

P1. 2.75p

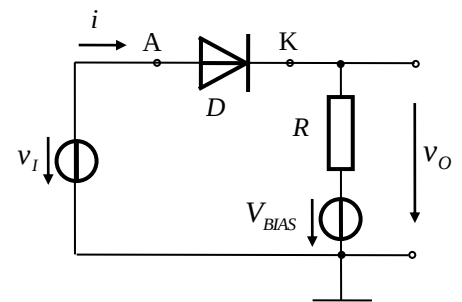
Consider $V_{BIAS} = -4V$.

a) 1p Deduce and plot the VTC $v_o(v_i)$ of the circuit, for D-ideal diode. Indicate the state of the diode and the expression of the output voltage for each region of VTC.

b) 0.5p Deduce and plot the VTC $v_o(v_i)$ considering for D the constant voltage drop model, with $v_D = 0.7V$.

c) 0.5p For $v_i(t) = 9\sin \omega t$ [V], plot $v_o(t)$.

d) 0.75p Consider another voltage source $v_{I2} = 5V$, and another diode D2 added to the circuit in the same way as v_i and D are connected (between ground and K point), $v_i(t) = 9\sin \omega t$ [V]. Draw the new circuit, find the expression of v_o and plot $v_o(t)$. What is the application of the new circuit?



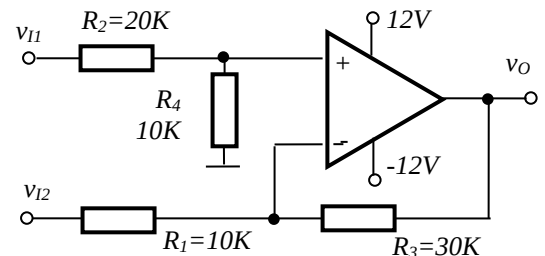
P2. 2.5p Assume OA – ideal.

a) 0.75p Find the expression of $v_o(v_{I1}, v_{I2})$, assuming O.A. in the active region. What is the application of the circuit?

b) 0.5p Plot $v_{I1}(t)$, $v_{I2}(t)$ and $v_o(t)$ for $v_{I1}(t) = 3\sin \omega t$ [V] and $v_{I2}(t) = 1 - 1\sin \omega t$ [V].

c) 0.5p Deduce the expressions for the input resistances (R_{i1} and R_{i2}) considering the input is only v_{I1} (with v_{I2} set to zero), respectively only v_{I2} (with v_{I1} set to zero), and the output resistance of the circuit, R_o .

d) 0.75p Find the new values for R_1 , R_2 , R_3 , R_4 in order to obtain: $v_o = 4(v_{I1} - v_{I2})$



P3. 2.75p

a) 0.5p At what end should be the tap of the potentiometer to obtain the maximum output voltage? Justify your answer.

b) 0.5p What is the adjustment range of V_o ?

c) 0.75p The maximum output current of the op amp, $I_{O,OA}$, is 20 mA. Modify the circuit so that the maximum output current of the voltage regulator will be 400mA.

d) 0.5p For $R_L = 10\Omega$, the tap of the potentiometer P in the middle and the maximum output current $I_{Omax} = 400mA$, compute the power dissipated by R_L .

